



HR WORKFORCE ATTRITION ANALYSIS

1. Introduction

Human Resource (HR) analytics plays a crucial role in understanding employee behavior and improving organizational performance. One of the major challenges faced by companies is employee attrition, which refers to employees leaving the organization voluntarily or involuntarily.

This project focuses on analyzing workforce attrition using HR data and identifying patterns that influence employee turnover. By leveraging Power BI, an interactive dashboard is created to visualize insights and support HR decision-making.

2. Problem Statement

High employee attrition can negatively impact business performance, increase hiring costs, and reduce productivity. Organizations need to identify the root causes of attrition and take proactive measures to retain employees.

3. Objectives

- Analyze employee attrition trends
 - Identify key factors affecting attrition
 - Build an interactive dashboard using Power BI
 - Provide actionable insights for HR teams
 - Suggest strategies to reduce employee turnover
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4. Dataset Description

The dataset contains detailed information about employees, including:

- Demographic details (Age, Gender, Education)
 - Job-related attributes (Department, Job Role)
 - Financial details (Monthly Income)
 - Work environment (OverTime, Work-Life Balance)
 - Performance metrics (Job Satisfaction)
 - Experience (Years at Company)
 - Attrition status
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5. Data Preprocessing (Power BI)

Data cleaning and transformation were performed using Power Query:

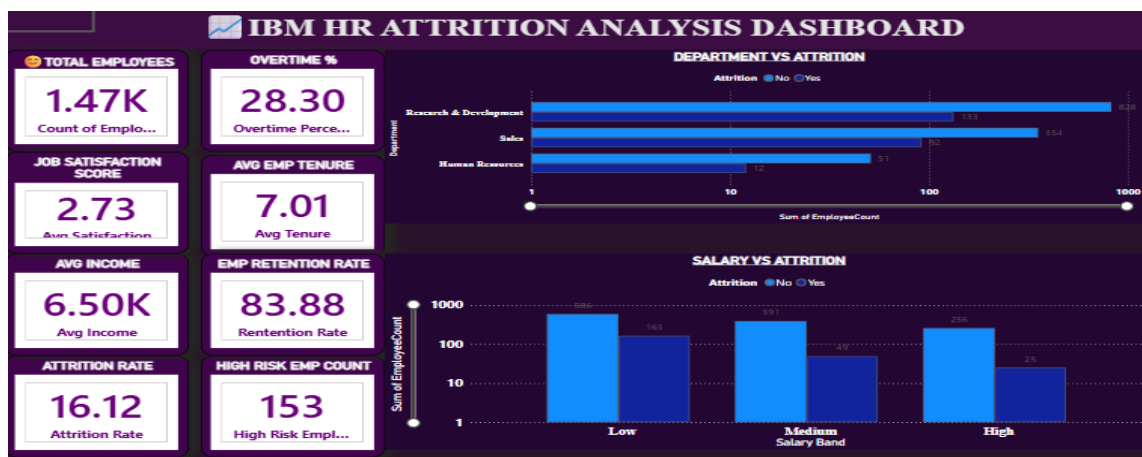
- Removed duplicate records
 - Handled missing/null values
 - Converted columns into appropriate data types
 - Created calculated columns:
 - Salary Band (Low, Medium, High)
 - Experience Level (0–5, 6–10, 10+ years)
 - Filtered unnecessary columns
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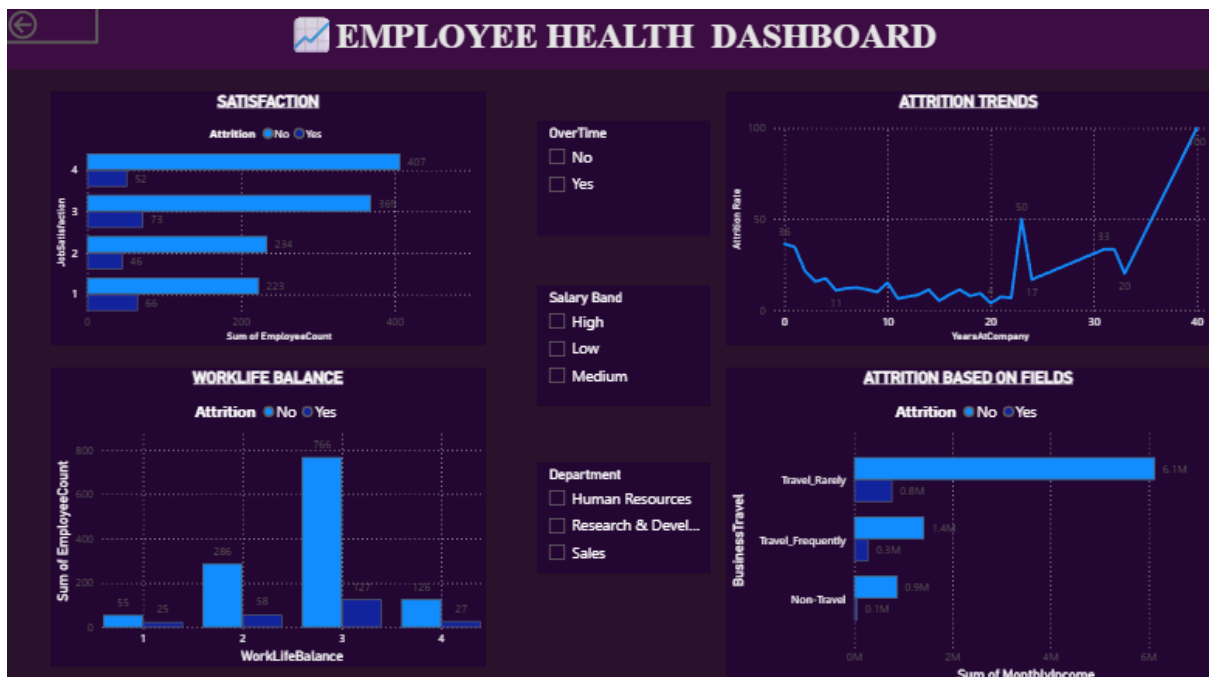
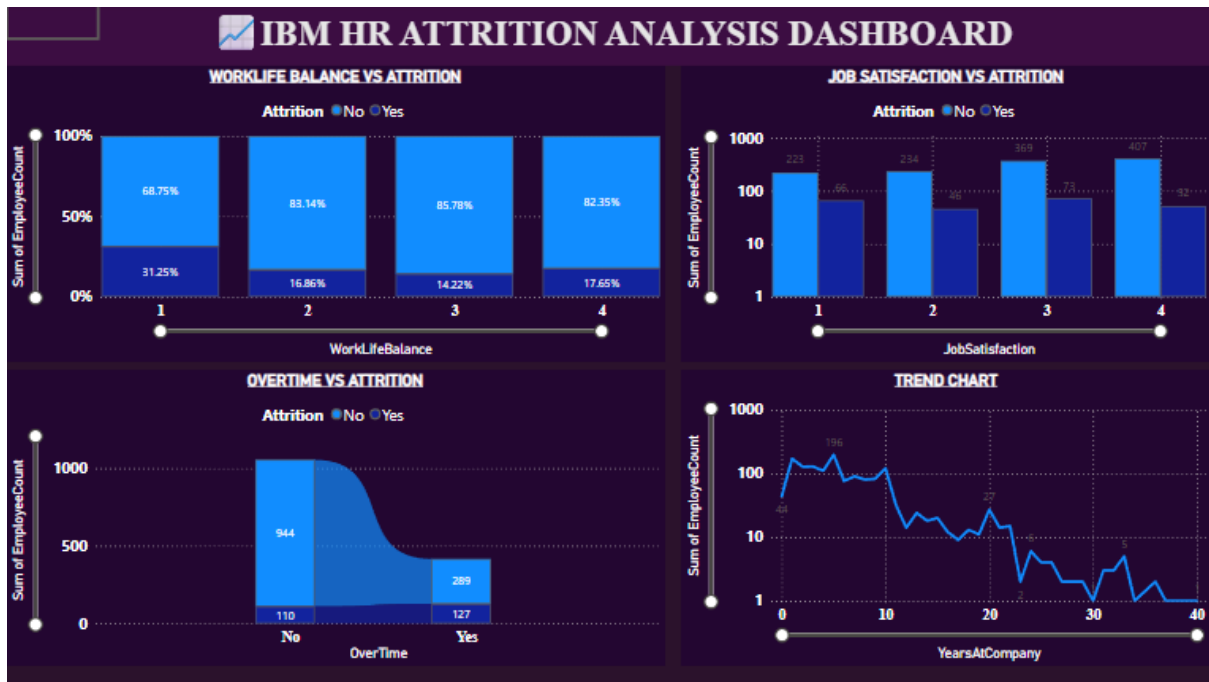
6. Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

- Total Employees
 - Attrition Count
 - Attrition Rate (%)
 - Retention Rate (%)
 - Average Monthly Income
 - Average Years at Company
 - Job Satisfaction Score
 - Overtime Percentage
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7. Dashboard Overview





The dashboard provides a comprehensive view of workforce attrition through multiple visualizations:

- Attrition by Department
- Attrition by Job Role
- Salary Band vs Attrition
- Overtime vs Attrition

- Job Satisfaction vs Attrition
 - Age Group vs Attrition
 - Experience vs Attrition
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8. Data Analysis

8.1 Attrition by Department

Certain departments such as Sales show higher attrition compared to others, indicating possible workload or pressure issues.

8.2 Salary Analysis

Employees in the lower salary band are more likely to leave the organization, highlighting the importance of competitive compensation.

8.3 Overtime Impact

Employees working overtime frequently show higher attrition, indicating poor work-life balance.

8.4 Job Satisfaction

Low job satisfaction strongly correlates with higher attrition rates.

8.5 Experience Analysis

Employees with 0–5 years of experience show the highest attrition, suggesting early career dissatisfaction or better opportunities elsewhere.

9. Key Insights

- High attrition observed in Sales department
 - Overtime is a major contributor to employee turnover
 - Low salary employees are more likely to leave
 - Job satisfaction plays a critical role in retention
 - Early-stage employees (0–5 years) are at higher risk
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10. Business Impact

- Increased recruitment and training costs
- Reduced team productivity
- Loss of experienced employees

- Negative impact on company culture
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11. Recommendations

11.1 Improve Work-Life Balance

- Limit excessive overtime
- Encourage flexible working hours

11.2 Compensation Strategy

- Review salary structures
- Provide performance-based incentives

11.3 Employee Engagement

- Conduct regular feedback sessions
- Recognize employee achievements

11.4 Career Growth Opportunities

- Provide training and development programs
- Offer clear promotion paths

11.5 Focus on High-Risk Employees

- Identify employees with low satisfaction
 - Implement retention strategies
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12. Conclusion

This project successfully analyzes HR workforce attrition using Power BI. The findings highlight that attrition is influenced by multiple factors such as salary, overtime, job satisfaction, and experience.

By addressing these factors, organizations can improve employee retention, enhance productivity, and build a stable workforce.

13. Future Scope

- Use Machine Learning models to predict attrition
- Implement real-time HR analytics dashboards
- Integrate employee feedback systems
- Perform sentiment analysis on employee reviews

